
A Math Confidence Reset Guide

Abel's Identity

The Wronskian Without Solving the Equation

For a second order linear homogeneous differential equation, the Wronskian of any two solutions is fixed in shape by a single coefficient, and the only freedom remaining is a scale constant. This guide builds that result from first principles, one transparent step at a time.

First Principles • Conceptual Understanding • Step by Step Transparency

Ordinary Differential Equations • Companion Reference

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1. Introduction to Abel's Identity

Abel's Identity, sometimes called Abel's formula, is a result that tells you the Wronskian of two solutions of a second order linear homogeneous differential equation. Its remarkable feature is what it does not require. You can write down the Wronskian without ever finding the solutions themselves.

Consider the standard form of the equation:

$$y'' + p(x)y' + q(x)y = 0.$$

If y_1 and y_2 are any two solutions, the Wronskian is the determinant

$$W(y_1, y_2)(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2.$$

The Wronskian is the quantity that measures whether two solutions are linearly independent. When it is nonzero, the two solutions form a genuine basis for every solution of the equation. When it is zero, they are linearly dependent and cannot span the full solution space.

Here is the conceptual heart of the matter. You might expect the Wronskian to depend on both coefficient functions, $p(x)$ and $q(x)$. Abel's Identity reveals that this is not the case. The entire behavior of the Wronskian is governed by $p(x)$ alone, the coefficient sitting in front of the first derivative. The coefficient $q(x)$ has no influence on it whatsoever.

Key Concept: What Abel's Identity Really Says

The shape of the Wronskian is dictated completely by $p(x)$. The only thing left undetermined is a single multiplicative constant, which is fixed once you know the Wronskian at one point. In symbols, Abel's Identity states:

$$W(x) = W(x_0) \exp\left(-\int_{x_0}^x p(z) dz\right).$$

This is the spirit of the Math Confidence Reset approach. Rather than memorizing the boxed formula as a magic incantation, you will see exactly why $q(x)$ drops out, why $p(x)$ is the only survivor, and why the constant turns out to be the initial value of the Wronskian. Once you have followed the logic, the formula stops being something to recall and becomes something you can rebuild on demand.

2. When to Use It

Abel's Identity is the optimal tool in a specific set of circumstances. Recognizing those circumstances is half the skill. Below are the scenarios where reaching for this formula is the efficient and natural choice.

2.1 You need the Wronskian but cannot solve the equation

Many second order linear equations, including those with variable coefficients, are difficult or impossible to solve in closed form. When you still need the Wronskian, for example to test linear

independence, Abel's Identity hands it to you using only $p(x)$. You never touch the solutions.

2.2 You want to know whether the Wronskian ever vanishes

Because the formula expresses the Wronskian as a constant times an exponential, and an exponential is never zero, the Wronskian is either identically zero or never zero on any interval where $p(x)$ is continuous. This single observation resolves a question about linear independence across an entire interval at once.

2.3 You are performing reduction of order

Suppose you already know one solution y_1 and you want a second, independent solution y_2 . Reduction of order requires the Wronskian as an intermediate quantity. Abel's Identity supplies it immediately, which lets you recover y_2 by a single integration. This is one of the most common practical uses of the formula.

2.4 The first derivative coefficient is simple, even if the rest is not

Because only $p(x)$ enters the formula, a complicated or even unknown $q(x)$ is irrelevant to the Wronskian. Whenever $p(x)$ is clean and easy to integrate, Abel's Identity is the fastest route, regardless of how unpleasant $q(x)$ may look.

A Quick Decision Rule

Ask yourself one question. Do I need the Wronskian, and is $p(x)$ easy to integrate? If the answer to both is yes, Abel's Identity is almost always the most efficient tool available.

3. How to Use It (Examples)

The two examples below apply the formula from first principles. The first treats constant coefficients, and the second treats variable coefficients. In each case we compute the Wronskian using Abel's Identity, then verify the result by hand to build trust in the method.

3.1 Example 1: A Constant Coefficient Equation

Problem. Find the Wronskian of two solutions of

$$y'' - 3y' + 2y = 0.$$

Worked Solution

Step 1. Identify $p(x)$ from standard form.

The equation is already in standard form, so we read off the coefficient of y' directly:

$$p(x) = -3.$$

Step 2. Apply Abel's Identity.

Substitute $p(x) = -3$ into the formula. The constant of integration is absorbed into the

leading constant C :

$$\begin{aligned} W(x) &= C \exp\left(-\int p(x) dx\right) \\ &= C \exp\left(-\int (-3) dx\right) \\ &= C \exp\left(\int 3 dx\right) \\ &= C e^{3x}. \end{aligned}$$

Step 3. Verify by computing the solutions directly.

The characteristic equation is $r^2 - 3r + 2 = 0$, which factors as $(r - 1)(r - 2) = 0$. The roots are $r = 1$ and $r = 2$, giving the two solutions

$$y_1 = e^x, \quad y_2 = e^{2x}.$$

Now compute the Wronskian by hand:

$$\begin{aligned} W &= y_1 y_2' - y_1' y_2 \\ &= e^x (2e^{2x}) - e^x (e^{2x}) \\ &= 2e^{3x} - e^{3x} \\ &= e^{3x}. \end{aligned}$$

This matches $W(x) = Ce^{3x}$ with $C = 1$, exactly as Abel's Identity predicted.

Takeaway. Abel's Identity delivered the form Ce^{3x} without any reference to the solutions. The direct calculation only confirmed what the formula already guaranteed.

3.2 Example 2: A Variable Coefficient Equation

Problem. Find the Wronskian of two solutions of the Cauchy and Euler equation

$$x^2 y'' - 2x y' + 2y = 0, \quad x > 0.$$

Worked Solution

Step 1. Convert to standard form.

Abel's Identity requires the leading coefficient to be 1. Divide every term by x^2 :

$$y'' - \frac{2}{x} y' + \frac{2}{x^2} y = 0.$$

Reading off the coefficient of y' gives

$$p(x) = -\frac{2}{x}.$$

Step 2. Apply Abel's Identity.

Substitute $p(x) = -2/x$ and integrate:

$$\begin{aligned}
 W(x) &= C \exp\left(-\int p(x) dx\right) \\
 &= C \exp\left(-\int \left(-\frac{2}{x}\right) dx\right) \\
 &= C \exp\left(2 \int \frac{1}{x} dx\right) \\
 &= C \exp(2 \ln x) \\
 &= C e^{\ln x^2} \\
 &= C x^2.
 \end{aligned}$$

Step 3. Verify by computing the solutions directly.

For a Cauchy and Euler equation, try $y = x^m$. Substituting gives the indicial equation $m(m-1) - 2m + 2 = 0$, which simplifies to $m^2 - 3m + 2 = 0$, with roots $m = 1$ and $m = 2$. The two solutions are

$$y_1 = x, \quad y_2 = x^2.$$

Now compute the Wronskian by hand:

$$\begin{aligned}
 W &= y_1 y_2' - y_1' y_2 \\
 &= x(2x) - (1)(x^2) \\
 &= 2x^2 - x^2 \\
 &= x^2.
 \end{aligned}$$

This matches $W(x) = Cx^2$ with $C = 1$, confirming Abel's Identity once again.

Takeaway. The coefficient $q(x) = 2/x^2$ never entered the calculation. Only $p(x)$ determined the Wronskian, which is precisely the power of the identity.

4. The Proof of Abel's Identity

Given a second order linear homogeneous ordinary differential equation of the form

$$y'' + p(x)y' + q(x)y = 0,$$

if y_1 and y_2 are two solutions to this equation, Abel's Identity provides a formula to compute their Wronskian, $W(y_1, y_2)(x)$, without needing explicit formulas for y_1 and y_2 .

4.1 Step 1: Setting up the System

Because y_1 and y_2 are solutions, they must individually satisfy the original differential equation:

$$y_1'' + p(x)y_1' + q(x)y_1 = 0, \tag{1}$$

$$y_2'' + p(x)y_2' + q(x)y_2 = 0. \tag{2}$$

To eliminate the $q(x)$ term, we cross multiply the equations by the opposite solution. We multiply the first equation by y_2 and the second equation by y_1 :

$$\begin{aligned}\text{Top: } & y_1'' y_2 + p(x) y_1' y_2 + q(x) y_1 y_2 = 0, \\ \text{Bottom: } & y_2'' y_1 + p(x) y_2' y_1 + q(x) y_1 y_2 = 0.\end{aligned}$$

4.2 Step 2: Elimination

We subtract the Top equation from the Bottom equation. The $q(x) y_1 y_2$ terms perfectly cancel out. Grouping the remaining terms and factoring out $p(x)$ yields:

$$(y_2'' y_1 - y_1'' y_2) + p(x) [y_2' y_1 - y_1' y_2] = 0.$$

4.3 Step 3: Introducing the Wronskian

The Wronskian of the two solutions is defined by the determinant

$$W(y_1, y_2) = (y_1)(y_2') - (y_1')(y_2).$$

This definition is exactly the expression inside the brackets of our subtracted equation. Next, we find the derivative of the Wronskian with respect to x using the product rule:

$$W' = (y_1' y_2' + y_1 y_2'') - (y_1' y_2' + y_1'' y_2).$$

The $y_1' y_2'$ terms cancel, leaving:

$$W' = y_1 y_2'' - y_1'' y_2.$$

This matches the first grouped term in our subtracted equation.

4.4 Step 4: Forming a First Order ODE

By substituting W and W' into the equation from Step 2, the complex second order system reduces to a separable first order linear ODE with respect to the Wronskian:

$$W' + p(x) W = 0.$$

4.5 Step 5: The Strategic Integrating Factor

To solve this first order ODE, we need an integrating factor, $\mu(x)$, that satisfies the condition that the derivative of μ with respect to x equals $p(x) \mu(x)$:

$$\frac{d\mu}{dx} = p(x) \mu(x).$$

While we could use an indefinite integral and solve for an arbitrary constant later, theoretical proofs require a generalized formula that mathematically pre solves the initial condition. By the Fundamental Theorem of Calculus, we can define an accumulation function using a definite integral from a fixed starting point ($x = 0$) to an arbitrary variable upper bound (x). We use z as a dummy variable of integration to avoid notation conflicts:

$$\mu(x) = \exp\left(\int_0^x p(z) dz\right).$$

This specific integrating factor has a crucial property. When evaluated at the starting point $x = 0$, the integral evaluates to zero, anchoring the integrating factor to exactly 1:

$$\mu(0) = \exp\left(\int_0^0 p(z) dz\right) = e^0 = 1.$$

4.6 Step 6: Reversing the Product Rule and Finding the Constant

Multiplying our first order ODE by this integrating factor allows us to condense the left side into the derivative of a single product:

$$\frac{d}{dx} [\mu(x) W(x)] = 0.$$

If the derivative of a function with respect to x is zero, the function itself must be equal to some constant C :

$$\mu(x) W(x) = C.$$

Because this equation must hold true for all values of x , we can determine the exact value of C by evaluating the function at our strategic anchor point, $x = 0$. The definite integral from 0 to 0 is 0, which forces the exponential term to vanish into 1:

$$\mu(0) W(0) = C \implies (1) W(0) = C \implies C = W(0).$$

By choosing the definite integral, the arbitrary constant is explicitly revealed to be the initial value of the Wronskian.

4.7 Step 7: The Final Formula

We substitute our known constant back into the equation:

$$\mu(x) W(x) = W(0).$$

Finally, we isolate $W(x)$ by dividing both sides by the exponential term, equivalently by multiplying by the negative exponent, to arrive at Abel's Identity:

Abel's Identity

$$W(x) = W(0) \exp\left(-\int_0^x p(z) dz\right).$$

More generally, anchoring at any convenient point x_0 rather than 0 gives the equivalent statement

$$W(x) = W(x_0) \exp\left(-\int_{x_0}^x p(z) dz\right),$$

and when only the form of the answer is needed, the indefinite shortcut $W(x) = C e^{-\int p(x) dx}$ captures the same content with the constant C left to be fixed by data.

Appendix A: The First Principles and the FTC

When solving a first order linear ordinary differential equation, the integrating factor method relies on finding a specific multiplier. To make the reverse product rule work, this multiplier must satisfy one strict mathematical condition. The derivative of μ with respect to x equals $p(x)\mu(x)$:

$$\frac{d\mu}{dx} = p(x)\mu(x).$$

To build a function that satisfies this condition, we look to the First Part of the Fundamental Theorem of Calculus (FTC). The FTC states that if you create an accumulation function using a definite integral from a constant lower bound to a variable upper bound, taking the derivative of that integral simply yields the original integrand evaluated at x . We use z as a dummy variable inside the integral so we do not confuse the variable being integrated with the upper limit x :

$$\frac{d}{dx} \int_{x_0}^x p(z) dz = p(x).$$

Why do this instead of just using an indefinite integral? An indefinite integral represents an infinite family of curves shifted by an arbitrary constant. However, to solve the differential equation, we do not need the whole family. We only need one valid multiplier. By using a definite integral from a constant lower bound to x , we are selecting one specific, anchored antiderivative.

Constructing the Integrating Factor

We hypothesize that our integrating factor takes the form of an exponential function raised to this definite integral:

$$\mu(x) = \exp\left(\int_{x_0}^x p(z) dz\right).$$

To prove this works, we take its derivative using the chain rule. The derivative of e^u is $e^u \cdot \frac{du}{dx}$:

$$\frac{d}{dx} \left[\exp\left(\int_{x_0}^x p(z) dz\right) \right] = \exp\left(\int_{x_0}^x p(z) dz\right) \cdot \frac{d}{dx} \left[\int_{x_0}^x p(z) dz \right].$$

Applying the FTC rule from above to that second term, we get:

$$\frac{d}{dx} \left[\exp\left(\int_{x_0}^x p(z) dz\right) \right] = \exp\left(\int_{x_0}^x p(z) dz\right) \cdot p(x).$$

Rearranging this slightly, it perfectly matches our initial requirement:

$$\frac{d\mu}{dx} = p(x)\mu(x).$$

Appendix B: Reversing the Product Rule and Finding the Constant

By multiplying the original differential equation by our specific integrating factor, this time choosing 0 as our constant lower bound so that $x_0 = 0$, the entire left side of the equation collapses via the reverse product rule into a single derivative set equal to zero:

$$\frac{d}{dx} \left[\exp \left(\int_0^x p(z) dz \right) W(x) \right] = 0.$$

If the rate of change, the derivative, of a function is exactly zero, the function itself must be a flat, unchanging constant. We will call it C :

$$\exp \left(\int_0^x p(z) dz \right) W(x) = C.$$

This equation is a universal truth for this system. It holds for any value of x . Because it is universally true, we can plug in any x value we want to solve for C .

By choosing to evaluate the function specifically at $x = 0$, we create a scenario where the integral's upper and lower bounds are identical:

$$\exp \left(\int_0^0 p(z) dz \right) W(0) = C.$$

In calculus, the accumulated area under a curve from a point to that exact same point is zero. Therefore, the integral evaluates to 0:

$$e^0 \cdot W(0) = C.$$

Since anything to the power of zero is 1, the exponential term vanishes completely:

$$1 \cdot W(0) = C,$$

$$C = W(0).$$

By anchoring our integrating factor with a definite integral starting at 0, the arbitrary constant C is instantly and cleanly revealed to be the initial value of the Wronskian at $x = 0$.

The Payoff

The strategic choice of a definite integral anchored at a known point does two jobs at once. It produces a valid integrating factor, and it pins down the constant of integration to the meaningful quantity $W(0)$ with no extra work. This is why the proof prefers the anchored definite integral over a generic indefinite one.